

BREECH-LOADING ARTILLERY

New materials began to be used to build artillery—muzzle-loaders as well as rarer breech-loaders—in the second half of the 19th century, revolutionizing artillery design. Cast-iron and bronze barrels were replaced by stronger ones of wrought iron and steel. There were also improvements in gunpowder manufacture which translated into longer range, more accuracy, and greater penetration. This was especially important in the days of the development of ironclad warships. Breech-loaders had always proved more practical than muzzle-loaders on ships (see p. 14). Breech-loading also meant that naval guns could now have long barrels, since it was no longer necessary to load at the muzzle, and this helped significantly to increase their range.

► ARMSTRONG RBL 12-POUNDER

Date 1859

Origin UK

Length (Barrel) 7ft (2.13m)

Caliber 7.62cm

Range 2 miles (3.1km)

This Armstrong rifled breech-loader, or RBL gun, required a crew of nine men to operate it. The gun that entered British Army service (shown here) in 1859 had a 7-ft (2.13-m) barrel, while the British Royal Navy used a 6-ft (1.83-m) barrel version. In 1863, the shorter version became standard.

▼ ARMSTRONG RBL 40-POUNDER

Date 1861

Origin UK

Length 9¾ft (3m)

Caliber 12cm

Range 1½ miles (2.5km)

The Armstrong 40-pounder was used by the British Royal Navy as a broadside gun (a gun used in a battery on one side of a ship), and by the army as a defensive gun in military forts. It saw action in the Royal Navy's bombardment of Kagoshima, Japan, in August 1863.

Trail

Carriage wheel

Rifled wrought-iron barrel

